

Exercise sheet 5

PRISM - Curves and Surfaces

1. Given two surface patches $\sigma_1 : U_1 \rightarrow \mathbb{R}^3$ and $\sigma_2 : U_2 \rightarrow \mathbb{R}^3$, whose images cover the same part of the a given surface S in $\mathbb{R}^3$, how would you relate them? In other words, define a “coordinate transformation” map to relate σ_1 and σ_2 just as we defined a reparametrization to relate two parametrizations θ_1 and θ_2 of curves.
2. Remember $\mathbf{n}(p)$, the unit normal to the surface at p , which is defined as $\frac{\sigma_x \times \sigma_y}{\|\sigma_x \times \sigma_y\|}$.
 - (a) Do you think it depends on the surface patch σ ? After all, if you change σ , the vectors σ_x and σ_y change.
 - (b) To find out the how $\mathbf{n}(\theta(t))$ changes along the parametrization θ , i.e. to know $\frac{d}{dt}\mathbf{n}(\theta(t))$, it is enough to figure out $\mathbf{n}_x$ and $\mathbf{n}_y$. Why is that enough?
 - (c) Show that $\mathbf{n}_x$ and $\mathbf{n}_y$ must be tangent to the surface. Therefore, they are of the form $\mathbf{n}_x = a\sigma_x + b\sigma_y$ and $\mathbf{n}_y = c\sigma_x + d\sigma_y$ for some scalars a, b, c , and d .
 - (d) Find out the scalars a, b, c , and d in terms of the components of the fundamental forms, E, F, G, L, M , and N . You might find the computation easier if you arrange the coefficients in a matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

You will then be able to express it as a matrix multiplication of two matrices which have E, F, G, L, M, N in their entries.

3. Consider the sphere of radius 1. Show that any curve that lies on the sphere and has geodesic curvature 0, must be a great circle. To do this, you must first define a suitable surface patch on part of the sphere.